

Smart City 2.0 — AI-Powered · Premium Course

2-day camp (5 hrs/day) · 12-15 yrs · from KES 9,000

Premium course outline **PREMIUM**

Overview

The ultimate crossover: build an Arduino smart city, then give it an AI brain.

DETAIL	
Track	Combo · Arduino + AI
Format	2-day camp (5 hrs/day)
Ages	12-15 yrs
Level	Advanced
Price	from KES 9,000
Maker Kit	Optional, KES 8,500 one-time: Arduino Uno R3 board + USB cable, Breadboard + 40 jumper wires, LEDs (red, yellow, green, white, RGB) + resistors, Push buttons, potentiometer and piezo buzzer, Light sensor (LDR) and sound sensor, HC-SR04 ultrasonic sensor + bracket, IR sensor, DHT11 temperature & humidity sensor, PIR motion sensor, SG90 servo motor, Relay module, 16×2 LCD screen (I2C), Battery holder, Storage box. Bought once and reused in every Arduino course. Kits are lent during sessions; buy one to keep building at home.
Standalone	No
Prerequisites	Any 2 free Arduino activities, Prompting power (free)

Programme

- Day 1 — Build.** Teams assemble Arduino smart-city modules (traffic, parking, street lights, environment sensors).
- Day 2 — Add Intelligence.** Use Teachable Machine to add a "vision" layer — a camera classifies traffic density and the AI recommends light timing. ChatGPT generates a city-planning report from sensor data. Teams present their AI-enhanced city to parents.

What they take home

 Their city module + an AI analysis report

Why parents love it

“Hardware meets AI. Two days, one smart city, built and brain-powered by kids.”

How to book

Places are limited and run on scheduled dates. To book or ask a question, email **tinkerwithanne@gmail.com** with the course name (**Smart City 2.0 — AI-Powered**), your child's age, and your preferred dates. Prices start from the amount above and may vary with group size and materials.